Project Plan

Document Number: 001

Date: Feb 6, 2026

Last updated: Feb 14, 2026

Allies Connect

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Georgia's Community Resource

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1. INTRODUCTION

Allies Connect will be a mobile-first web platform designed to easily connect Georgia residents with nearby and available community resources at the tap of a screen. These resources will include food distribution sites, community fridges, emergency pop-ups, shelters, nonprofit events, and volunteer opportunities throughout the state of Georgia. Through this unified interface, the goal is to increase accessibility for underserved people and improve engagement between volunteers and service providers.

The motivation for this project originates from the increased dependence on nonprofit organizations for essential services and lack of a centralized platform for easy accessibility to these organizations' information. Currently, information is scattered across various nonprofit organizations' websites, social media accounts, and flyers, making it difficult for people to access all the available information about events and opportunities. Allies Connect will seek to address this issue by creating an all-encompassing platform that provides Georgia residents with readily available and up-to-date information about community outreach and nonprofit operations. The system will be designed not only for immediate usage, but will act as a foundation for future improvements, including various automated features to minimize the need for human involvement.

This Project Plan describes the scope, context, stakeholders, activities, resources, schedule, constraints, risks, quality goals, and processes of Allies Connect. It is intended to provide a guide to how development work will be managed throughout the project lifecycle. It also established a baseline for how project progress and success will be measured.

2. PROJECT OVERVIEW

2.1 Scope

2.1.1 Identification

This document is laid out for the system titled "Georgia's Community Resource - Allies Connect," also referred to by the shorthand, "Allies Connect."

System Name: Allies Connect
Project Phase: Phase I - Manual-First Minimum Viable Product (MVP) + Automation Feasibility Research
Version: 1.0.0
Release: Academic Prototype Release (1.0)

2.1.2 System Overview

Allies Connect will be a web-based, mobile-first platform designed to enhance accessibility to all community outreach and nonprofit organizations within the state of Georgia. It will enable residents to easily find information relating to events while also giving nonprofit organizations state-of-the-art tools to manage events, information, and volunteer personnel.

The system is being developed as part of a Master of Science in Software Engineering (MSSWE)-sponsored academic capstone project at Kennesaw State University for Allies Way, Inc. The project's development is led by students under faculty supervision, with weekly meetings to obtain sponsor feedback. The project's platform will be hosted on a KSU-provided infrastructure (Linux Server) and will initially operate as a functional prototype. The system is intended to operate within the state of Georgia, accessible via mobile or desktop web browsers. Key stakeholders for this project include:

Sponsor: Allies Way, Inc.
Acquirer: Allies Way, Inc.
Users: Georgia residents, nonprofit organizations, administrators, volunteers
Developers: Student capstone team

Support: KSU faculty and infrastructure

Relevant documents include the Capstone Project Specification document titled "16-Georgias Community Resource-Allies Connect" and weeks one through four Progress Report documents.

2.1.3 Document Overview

This Project Plan document defines the development and management framework for the Allies Connect project. The scope, objectives, stakeholders, development activities, schedule, constraints, risks, and quality goals can be found within this document. It serves as a guide for the development team, sponsors, and Kennesaw State University faculty to ensure the project is completed with an acceptable level of quality and within hard time bounds.

2.2 Background, Objectives, and Scope

Many Georgia residents rely on nonprofit and community outreach organizations for food, shelter, crisis assistance, and community support. It is hard for people to access the information for these things because it is typically scattered, outdated, or manually maintained across platforms like websites, social media, static documents, phone trees, and flyers. Nonprofits typically struggle with keeping information updated, managing volunteers, publishing events consistently, and reaching underserved communities.

The key objectives of Allies Connect are to create a centralized platform for access to community resources across Georgia, reduce administrative burden on non profits, improve accessibility for people in need and potential volunteers, and provide a foundation for future automation capabilities. A core goal for the user experience (UX) is to provide a calm, simple, and easy-to-use interface for people who may be under stress.

The scope of this project includes delivering a manual-first, fully functional MVP with core features including resource listings, events, volunteer management, a provider portal, and admin moderation, while also identifying potential automation enhancements for future development.

2.3 Operational Policies and Constraints

The following policies and constraints control the development of the Allies Connect project. The platform must be designed to be web-based and mobile-first, not native mobile. The backend must be hosted on KSU-provided infrastructure and/or some other no-cost environment. Accessibility should be considered using Web Content Accessibility Guidelines (WCAG) basics. The system must be modular enough for the future Phase II expansion. All application programming interfaces (APIs) must be no-cost unless provided by the sponsor. Only automated features approved by the sponsor will be implemented in this project.

The system design will keep in mind that not all users will be computer-literate, and that they may have devices with small screens, slow and/or intermittent Internet connectivity, and older hardware. Language used in the user interface will assume lower literacy levels to be inclusive of all users.

2.4 Users or Involved Personnel

Public users are Georgia residents who will be one of the primary users of this system. They will access the platform when seeking food assistance, shelter, events, or volunteer opportunities. These users will primarily interact through searches, filtering, and sign-up forms. They will require an easily navigable, simple, and intuitive interface. Service providers will be nonprofit organizations, non-governmental organizations (NGOs), churches, and other community outreach programs responsible for creating and maintaining resource listings, events, and volunteer opportunities. Organizational users will require authenticated access, data management tools, and exportable volunteer data. Administrators will be the system's moderators and will manage its data and integrity. They will be responsible for approving providers, moderating content, removing outdated or false listings, and monitoring logs.

The development team is the group of KSU students responsible for designing, testing, implementing, documenting, and delivering the final Phase I system to the sponsor by using their knowledge obtained throughout their time studying at KSU. Allies Way is the sponsor for this project and will be responsible for providing the

direction of the project and constructive feedback to the development team. The KSU faculty are responsible for providing technological insight, infrastructure, and grading to the project team.

3. DEVELOPMENT BACKGROUND/APPROACH

Allies Connect will be developed as a web-based, mobile-first application that can be accessed on desktop and mobile web browsers. A milestone-driven approach will be used for development. The development activity will support the three major milestones that correspond to required deliverables and graded evaluations. Requirements will be verified early to confirm their accuracy and to align the project objectives with the sponsor's expectations. Functionality will be delivered incrementally so that certain features can be completed and reviewed early, including the use of AI-generated, throwaway prototypes. This allows for ongoing feedback from the sponsor and the ability to make timely adjustments as the project progresses.

A modern full-stack web development environment will be used for development. This includes React for the frontend, Node.js with the Express framework for the backend, MySQL for database managements, Google Maps for location features, and a free-tier email service API. The Allies Connect system will be primarily hosted on a KSU Linux environment. Standard tools and version practices will be used for development and testing to support collaboration and maintainability. More information is available on our versioning strategy in Project Configuration and Data Management, later in this document.

The development team consists of five graduate software engineering students who are skilled in full stack development, UX design, database management, and research. During Milestone 1, the team will meet with the project sponsor weekly to gather data, clarify requirements, and discuss design directions and feature priorities. Bi-weekly meetings will be held during Milestones 2 and 3 to discuss progress, get feedback, and make adjustments. The team will have internal meetings weekly to coordinate and track tasks.

3.1 High Level Estimates

This project will be completed in 15 weeks. Each member of the development team is expected to contribute 12-15 hours per week, resulting in an estimated 180-225 hours per team member. The development team includes five team members, so the total estimated effort is 900-1125 person hours.

High-level estimates by milestone are shown in the table below.

Phase/ Milestone	Estimated Effort (Person-hours)	Estimated Lines of Code	Estimated Documentation Pages	Estimated Test Cases	Duration (Weeks)
Milestone 1: Requirements & Foundation	240-300	0	30-40	0	4
Milestone 2: Prototype & Core Features	360-450	2,000 - 3,000	20-30	15-20	6
Milestone 3: Final Integration & Delivery	300-375	1,000 - 2,000	10-20	20-25	5
Estimated Total	**900-1125 person-hours**	**3,000 - 5,000 LOC**	**60-90 pages**	**35-45**	**15 weeks**

3.2 Key Contacts and Stakeholders

Role	Name	Email	Description
Project Sponsor	Allies Way, Inc. Dr. Nytoyia Laughlin	nlaughlin@alliesway.org	The project sponsor provides information about the requirements for the project, evaluates interim deliverables, and signs off on the final deliverable.
Development Team Leader	Molly Calhoun	mcalho21@students.kennesaw.edu	The development team leader gathers information from other developers to provide this information to other stakeholders on a regular basis. She also leads meetings with the client.
Development Team	Takeshia Banks David Castro Tarik Davis Ryan Hanrahan	tbanks70@students.kennesaw.edu dcastr19@students.kennesaw.edu tdavi192@students.kennesaw.edu rhanrah1@students.kennesaw.edu	The development team coordinates the individual tasks so that all work is performed in coordination and is completed on time.
Instructor	Dr. Reza M. Parizi	rparizi1@kennesaw.edu	The instructor creates the original high-level requirements document, evaluates milestone deliverables, and provides guidance throughout the project.
Capstone Faculty	Alla Kemelmahker Taylor Cuffie	akemelma@kennesaw.edu tcuffie1@kennesaw.edu	Capstone faculty attend client meetings to ensure they stay on track, and participate in developer-client communications to ensure they are serving the project well.

4. FEATURES, PRIMARY DELIVERABLES, AND EXTERNAL COMMITMENTS

4.1 Feature List

1. Public Resource Finder
    - The system will include a resource details page with a mobile-friendly map that allows filtering data by category, ZIP code, and date. Users will be able to manually enter ZIP code or use geolocation to find resources.

2. Events Directory
    - The system will show a centralized directory of events in calendar and list views. Service providers will be able to manually create and manage events. Public users will be able to filter events by date, ZIP code, and category.

3. Volunteer Management System
    - Service providers will be able to create and manage volunteer opportunities. Authenticated (logged-in) public users will be able to sign up for volunteer opportunities. Service providers will be able to view, manage, and export volunteer sign-up information.
    - If time allows, a feature to enable automated email reminders to be sent to volunteers will be implemented.

4. Organization Portal
    - Portal for authenticated service providers that allows them to manage their organizational profiles, resources, events, and volunteer postings.

5. Administration Console
    - Console for system administrator to access management tools for approving or rejecting service providers accounts, managing listings, and viewing system logs.
    - A main component of the system administration console will be to confirm the authority of accounts creating organization records in the system. Individuals will be able to create an organizational account and set up an organization's account in the system. That organization will be visible in any and all listings of organizations. However, until the administrators validate that the account that created the organization record is an authorized representative of the organization, they will not be able to create events or volunteer opportunities in the system. The UI will inform them that their account is "pending" if they attempt to create one of these records. An email will be sent out once they have been moved from "pending" to "confirmed."

6. Integrated UI
    - While the Events Directory, Volunteer Management System, and Organization Portal are all discrete elements of the application, visually they will be integrated. Users of the site will be able to seamlessly move from an Event to the Organization that is holding it, to the Volunteer opportunities offered by that Event.
7. Activity Logging
    - The system will record basic logging of key actions to support accountability and system maintenance.

8. Automation Feasibility Research
    - The project will include research and documentation that evaluates the feasibility of future automation features.

4.2 Customer Deliverables

The following deliverables will be provided to the project sponsor throughout the development process.

Date	Deliverable	Description
Week 4	Project Plan	Approved plan that describes the scope, development

		approach, features, schedule, and stakeholders.
Week 4	Requirements Document & Architecture Diagrams	Requirements document, user personas, use cases, user stories, architecture diagrams, entity-relationship diagram (ERD), user interface (UI) wireframes, automation research plan
Week 5	Milestone 1 Presentation	Presentation summarizing requirements and foundation
Week 10	Milestone 2 Presentation	Presentation showing prototype of core features
Week 14	Milestone 3 Presentation	Presentation of completed system and demo
Week 15	Functional MVP	Working manual-first web platform with the defined core features.
Week 15	System Package	Complete source code, technical documentation, test report, setup/installation guide
Week 15	Research Report	Report on automation feasibility
Week 15	Future Automation Roadmap	Roadmap for implementing automation in Phase II
Week 15	Project Acceptance Document	Document signed by sponsor to confirm receipt of deliverables

5. PROJECT SCHEDULE

Date (YYYY-MM-DD)	Milestone/ task	Deliverable	Remarks
2026-02-16	Milestone 1 - Requirements and Foundation	Full requirements document and project plan, system architecture design layout, and research plan for system automation feasibility	
2026-03-26	Milestone 2 - Core Functionality and Project Prototype	Backend API and Database; login + dashboard; resource, volunteer, and event creation; public resource finder; events calendar; volunteer signup system; Admin	

Date (YYYY-MM-DD)	Milestone/ task	Deliverable	Remarks
		Approval Workflow; Logging; documentation	
2026-04-26	Milestone 3 - Final Integrated Prototype	Full web platform functionality, research report, system demo, test cases and QA report, final presentation uploaded to GitHub, complete GitHub repo with documentation	

6. DEPENDENCIES AND CONSTRAINTS

6.1 Constraints

6.1.1 Time

The team will have 10 full weeks to bring the project to fruition after the requirements gathering phase. At the five week mark our team will be expected to have the core functionality of the site completed, and the rest of the five weeks until project completion will be spent on refinement, and research for the automation systems.

6.1.2 Budget

Since our team is not receiving any funding for this project, all resources (including site hosting and database management) we will need to find creative solutions to deliver on the project requested of us in the spec sheet. One of our biggest requests is the ability of the system to log certain actions by users, and generate forms. Those forms require a lot of data to synthesize. We'll need to be careful about what data we're storing, and only store information that is absolutely vital to do so.

6.1.3 Scope

To develop this web application we need to develop three distinct systems that all interact with each other. First is a service to create entities and store them with location and time sensitive information. This will allow us to create an interactive map where users can view and create resources, events, and volunteer opportunities. These entities will need to be able to store data, to then be able to provide to users on request. Next is the calendar system, which will help store and display information about events, and the events users are signed up for. Finally, we will create a volunteer management system, which will allow volunteers to sign up for specific events and opportunities, and allow resources to manage shifts for volunteers to sign up for. The interplay of these connected systems should provide a strong enough foundation for us to create the rest of the application around.

6.2 Dependencies

To begin development on our project, the most critical component will be creating the form builders that will allow us to dynamically construct all forms from the resource information pages to the intake forms for volunteer roles. Once those are complete we'll be able to flesh out the structure of how events, resources, and volunteer positions are structured. Concurrently with that fleshing-out process, we will be able to work on the mapping tools and the calendar tools. With all of those components in place, time allowing, we will then be able to construct the volunteer management system by integrating the other components we've already built. Tasks that can be completed at any time during development are: the construction of the database, the API endpoints, and the automation research that will be completed for our final deliverable. If time runs short, the volunteer management system will be created as a fully-functional but not-feature-complete implementation.

7. PROJECT CONFIGURATION AND DATA MANAGEMENT

7.1 Configuration Management

7.1.1 Version Control

The team will use GitHub and Git as our version control for software code. We will use Google Drive and Google Docs for our documentation organization and version control.

7.1.2 Branching

The team will use the GitFlow model for branch management. This may be a bit overkill for such a short-duration project, especially since we're planning a pretty segmented workload (with it being rare for more than one person to be working in the same area of the code). However, it isn't a heavy overhead, so it seems like the best option.

7.2 Tools and Infrastructure

7.2.1 Build Automation

Because this project stack will use React, which doesn't need to be compiled to run on the server, no build automation is planned.

7.2.2 Data Schema Management

Data schemata will be stored in a regularly exported Data Definition Language (DDL) file. That file will be included in the Git repository.

7.3 Data Management

7.3.1 Environment Configuration

Secrets, such as API keys and passwords, will be stored in `.env` files outside of the code repository. These will be incorporated into the server using the Dotenv library.

7.3.2 Documentation Management

Documentation will be stored in a combination of Google Docs files and Markdown files. Documents like this one will be stored in Google Docs, while documentation more geared for users of the system, like installation guides and user guides, will be stored in the Git repository in Markdown format.

8. PROJECT PROCESS

8.1 Software Life Cycle Model

Due to the amount of work that we'll need to complete in a short amount of time, our team will be following an AGILE development process. Each week will be a sprint, dividing up the work into manageable sections that we will be expected to develop, push to the main branch, and fix any merge conflicts that arise. At the beginning of next week's sprint we will assess the material that was covered in the previous week's sprint, and reevaluate the material we plan to cover in the next week.

9. GLOSSARY

Acronym or term	Spelled out acronym or term description
AGILE	The AGILE development process/Software Life Cycle Model

APIs	Application Programming Interfaces
DDL	Data Definition Language
ERD	Entity-Relationship Diagram
KSU	Kennesaw State University
Manual-first	This is a reference to the initial implementation of the app, where all actions are explicitly performed by a human. Later iterations may involve automation to perform some actions, such as importing data.
MVP	Minimum Viable Product
MSSWE	Master of Science in Software Engineering
NGOs	Non-governmental organizations
UI	User Interface
WCAG	Web Content Accessibility Guidelines

10. CHANGE RECORD

Changes made Feb 14, 2026 :

- 2.2.: Updated the section to ensure that the non-functional goal of a calm interface is captured.
- 2.3.: Updated the section to reflect the potential user limitations we are ensuring we accommodate.
- 4.1.5.: Added an entry describing the constraints on a newly-created organizational record.
- 4.1.6.: Added an entry to clarify that the user experience will be an integrated, seamless experience rather than feeling like three discrete elements.
- 6.2.: Updated the section to clarify that the volunteer management system will be pared down to a functional but not-feature-complete implementation if time runs short.